

HARTSVILLE, SC, USA

WMO: 720633

Lat: 34.400N

Lon: 80.117W

Elev: 365

StdP: 14.50

Time Zone: -5.00 (NAE)

Period: 09-19

WBAN: 00219

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 23.0	(c) 27.6	(d) 2.7	(e) 6.4	(f) 29.8	(g) 8.9	(h) 8.8	(i) 34.7	(j) 16.3	(k) 54.1	(l) 14.5	(m) 52.7	(n) 3.0	(o) 10	(p) 0.357

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.8	95.4	73.3	93.1	73.1	90.9	73.2	79.4	85.0	78.2	85.1	77.2	85.1	5.3	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.6	150.3	81.1	76.9	142.2	80.5	75.0	133.0	79.5	43.0	85.4	41.8	86.8	41.0	82.1	84.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.7	12.3	10.8	17.2	99.0	4.9	3.1	13.7	101.2	10.8	103.0	8.0	104.8	4.5	107.0
			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	63.6	44.6	48.6	54.0	63.4	71.7	78.5	80.7	79.4	75.5	64.0	53.3	48.4
	DBStd	14.76	9.94	9.65	9.48	7.49	5.89	4.07	3.00	3.30	4.88	7.72	7.56	9.29
	HDD50	624	225	133	67	5	0	0	0	0	0	2	48	144
	HDD65	2579	633	464	359	116	18	0	0	0	2	111	357	519
	CDD50	5586	58	93	193	408	672	854	951	912	764	437	148	96
	CDD65	2068	2	5	20	70	225	404	486	447	317	81	6	5
	CDH74	18754	12	38	194	731	1966	3896	4705	3957	2595	614	35	11
	CDH80	8088	0	1	30	190	740	1841	2261	1794	1080	150	1	0
	WSAvg	4.0	4.0	4.6	4.8	5.0	4.0	3.4	3.3	3.1	4.0	4.0	4.0	3.9
	PrecAvg	45.1	3.4	3.4	3.5	3.0	3.1	4.8	5.3	5.3	4.7	3.1	2.8	3.3
(15) Precipitation	PrecMax	53.1	7.6	6.9	7.2	5.6	5.7	8.9	8.3	9.1	10.1	5.8	6.5	7.9
	PrecMin	30.3	1.0	1.3	0.4	0.9	1.4	2.0	2.0	1.9	0.6	0.0	0.5	0.8
	PrecStd	5.6	1.9	1.6	1.6	1.4	1.1	1.7	2.0	2.4	2.6	1.6	1.5	1.9
	PrecStd	5.6	1.9	1.6	1.6	1.4	1.1	1.7	2.0	2.4	2.6	1.6	1.5	1.9
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	75.3	79.0	83.5	88.2	94.5	99.3	99.0	96.9	94.8	89.7	78.9	75.1	
	0.4% MCWB	62.9	64.2	62.9	65.1	70.8	74.2	74.3	71.7	70.0	73.1	66.1	67.0	
	2% DB	69.7	73.0	79.4	85.7	90.4	95.2	95.4	93.3	91.2	83.7	73.0	71.7	
	2% MCWB	59.9	61.4	61.9	66.2	69.4	74.1	74.4	73.2	72.0	71.1	61.9	65.6	
	5% DB	64.1	69.7	75.0	81.7	88.0	91.3	92.8	90.9	90.0	80.8	70.3	68.0	
	5% MCWB	57.4	59.5	60.2	64.6	69.5	73.3	74.3	73.7	71.4	68.2	61.6	63.2	
	10% DB	60.9	64.0	71.6	78.8	84.1	90.0	90.5	89.8	87.6	77.1	67.6	63.6	
	10% MCWB	54.0	56.3	58.7	63.1	69.2	72.9	74.6	73.8	71.7	66.5	59.8	59.4	
	0.4% WB	66.4	66.5	67.1	72.8	76.0	79.4	81.4	81.1	78.8	76.8	69.5	69.8	
	0.4% MCDB	71.5	73.2	74.8	78.2	81.9	85.9	87.1	86.0	84.0	82.1	71.3	72.0	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	63.6	64.5	65.2	70.9	73.6	77.9	79.5	79.2	76.6	74.9	66.7	67.2	
	2% MCDB	66.5	70.2	74.5	77.3	81.0	86.3	85.0	84.9	84.3	80.9	71.1	70.3	
	5% WB	59.4	62.2	63.5	68.5	72.6	76.7	78.3	78.0	75.0	72.9	64.1	63.6	
	5% MCDB	62.5	67.9	69.7	75.5	80.4	85.9	85.5	84.5	83.1	77.3	67.5	64.9	
	10% WB	55.3	58.7	61.5	66.1	71.6	75.4	77.3	76.9	73.8	70.2	61.8	60.4	
	10% MCDB	58.2	63.8	68.4	75.3	79.8	84.9	85.7	84.3	81.5	73.5	65.3	63.4	
(35) Mean Daily Temperature Range	MDBR	19.6	20.7	22.6	24.3	21.9	21.8	20.8	19.5	20.3	21.2	21.6	18.4	
	5% DB MCDBR	24.4	24.3	27.5	28.5	26.8	25.0	24.1	22.5	24.4	24.0	24.3	19.9	
	5% DB MCWBR	14.9	14.7	12.7	11.7	9.1	7.8	8.0	7.9	8.3	10.7	14.6	13.2	
	5% WB MCDBR	19.9	22.3	22.9	21.3	20.7	22.2	20.8	18.9	19.6	18.1	19.5	17.0	
	5% WB MCWBR	16.0	15.9	12.5	11.7	8.6	8.5	8.0	7.4	7.9	10.2	13.8	14.5	
	5% WB MCWBR	16.0	15.9	12.5	11.7	8.6	8.5	8.0	7.4	7.9	10.2	13.8	14.5	
(40) Clear-Sky Solar Irradiance	taub	0.314	0.324	0.358	0.405	0.465	0.491	0.542	0.513	0.431	0.387	0.336	0.310	
	taud	2.507	2.477	2.409	2.267	2.151	2.117	1.999	2.070	2.286	2.381	2.481	2.554	
	Ebn at Noon	285	293	290	280	263	255	241	246	263	267	273	280	
	Edh at Noon	26	30	35	43	49	50	56	51	40	33	27	24	
(44) All-Sky Solar Radiation	RadAvg	866	1069	1416	1776	1942	2032	1928	1750	1502	1243	951	757	
	RadStd	46	114	108	128	156	140	117	82	127	150	77	67	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (69 neighbors)	+0.94	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+304	+220

Nomenclature: See separate page